

Malong Dzhangu SQL Fundamentals

1. Selecting & Sorting

a. Retrieve all columns from the employees table

SQL Syntax: SELECT *

FROM employees;

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape-Town
3	Cathy	Zulu	Finance	65000	2017-07-19	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Preterial
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

b. Find all unique departments

SQL Syntax: SELECT DISTINCT department

FROM employees;

department
IT
HR
Finance
Marketing

c. Retrieve first and last names ordered by salary descending

SQL Syntax: SELECT first_name,

last_name

FROM employees

ORDER BY salary DESC;

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

d. Retrieve the top 3 highest-paid employees

SQL statement: SELECT id,
first_name,
last_name,
Salary
FROM employees
ORDER BY Salary DESC
LIMIT 3;

id	first_name	last_name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

2. Filtering and WHERE

e. Find employees in the IT department

SQL syntax: SELECT id,

first-name,

last-name,

department

FROM employees

WHERE department = 'IT';

id	first-name	last-name	department
1	Alice	Green	IT
8	Eva	Naidoo	IT

f. Find employees in finance with salary > 60000

SQL syntax: SELECT id,

first-name,

last-name,

department,

salary

FROM employees

WHERE department = 'finance'

AND salary > 60000

g. Find employees in HR or Marketing

SQL syntax: SELECT id,

first-name,

last-name,

department

FROM employees

WHERE department = 'HR'

OR department = 'Marketing'

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

4. Find employees not in IT

SQL syntax SELECT id,

first-name,

last-name,

department

FROM employees

WHERE department $\neq$ 'IT';

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

5. Find employees in IT, HR, or Finance using IN

SQL syntax SELECT id,

first-name,

last-name,

department

FROM employees

WHERE department IN ('IT', 'HR', 'Finance');

id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

1. Find employees in IT with salary > 65000 and city Johannesburg

SQL syntax SELECT id,

first-name,

last-name,

department,

salary,

city

FROM employees

AND salary > 65000

AND city = 'Johannesburg';

id	first-name	last-name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg